

AAYUSH PURI

AI&MachineLearningEngineer

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PROFESSIONAL SUMMARY

Experienced AI & ML Engineer with 9 months of hands-on experience in designing, developing, and deploying machine learning and deep learning solutions. Proven expertise in computer vision, natural language processing, and model optimization for real-world applications. Strong proficiency in Python, PyTorch, TensorFlow, and deployment on edge devices including Jetson, TensorRT, and DeepStream.

TECHNICAL SKILLS

Programming: Python, SQL

ML/DL: PyTorch, TensorFlow

Computer Vision: OpenCV, YOLO

NLP & AI: Whisper, LLMs, VLMs, Generative AI

Data Processing: NumPy, Pandas, Matplotlib

Deployment: ONNX, TensorRT, DeepStream, Docker

Cloud & DevOps: Git

Specializations: Stereo Depth Estimation, Model Optimization

EDUCATION

Bachelor of Technology - Computer Science

Dr. A.P.J. Abdul Kalam Technical University

2025 | Specialization: AI & ML

CERTIFICATIONS

✓ AI & ML Engineering - Enord (2024)

PROFESSIONAL EXPERIENCE

AI & Machine Learning Engineer

Enord – Delhi, India | December 2024 – Present (9 months)

- Designed deep learning pipelines for real-time drone-based aerial video analysis and autonomous navigation systems
- Developed and optimized YOLO models for crowd detection, achieving 30% improvement in inference speed through TensorRT
- Built stereo depth estimation models (ESM Stereo, RealSense) for obstacle detection and 3D mapping
- Implemented LLMs for natural language drone command processing and control systems
- Integrated Whisper speech-to-text and VLMs for intuitive drone control interfaces
- Deployed ML models into ROS2-based systems for real-time edge inference on Jetson AGX Orin
- Optimized training workflows for 24K+ dataset images, improving pipeline efficiency by 25%

Key Achievements:

- 90% accuracy improvement in crowd detection on drone footage
- 35% reduction in model inference latency through optimization
- Successfully deployed production-ready AI models on edge devices

KEY PROJECTS

Custom Whisper for Hinglish Recognition Fine-tuned Whisper model for Hinglish dataset, achieving 85% accuracy on mixed-language speech-to-text transcription. Deployed for real-time drone command systems.

YOLOv11m Multi-Class Detection Pipeline

Trained custom YOLOv11m on 24K+ images for people, bird, and FOD detection. Achieved mAP@0.5 of 0.92 and deployed for real-time crowd monitoring.

Drone Vision AI Platform

Built comprehensive real-time system integrating object detection, tracking, stereo depth estimation.

Stereo Depth Estimation System

Developed stereo vision system for accurate depth estimation and 3D mapping, optimized for real-time performance on Jetson AGX Orin with sub-centimeter accuracy.